MICROECONOMICS

MICRO-07

Time: 5 minutes

Outcome: Calculate Cross-Price Elasticity of Demand.

Difficulty: ●●○ Intermediate

Cross-Price Elasticity of Demand

THE CORE IDEA

Cross-price elasticity measures how demand for one good responds to a price change in another good. A large positive cross-price coefficient signals a strong substitution relationship. Complementary goods connect through opposite movements between one price and the other good's demand. Demand for one good moving in the same direction as the other good’s price implies positive cross-price elasticity. Little demand response to the other good’s price implies little economic relationship between the goods.

HOW TO RECOGNIZE IT

- The price of butter rises and demand for margarine rises.
- Little demand response to the other good’s price implies little economic relationship between the goods.
- Cross-price elasticity between printers and ink cartridges is -0.7.
- A negative cross-price coefficient means demand moves opposite the other good’s price.
- The price of Product Y falls 8 percent.

WATCH OUT

Do not confuse cross price with income or own price. A negative cross-price elasticity indicates complements, so higher printer prices reduce demand for ink.

WORKED EXAMPLE: IDENTIFYING SUBSTITUTES

The price of printers rises by 10%, and demand for ink cartridges falls by 7%. Cross-price elasticity is -7% / 10% = -0.7. The negative sign indicates that printers and ink cartridges are complements because a higher price for printers reduces demand for ink.

CHECK YOURSELF

The price of Product Y rises 10 percent. Cross-price elasticity of demand for Product X is −0.5. Demand for X is predicted to:?

READY? Return to the game and master this concept.